Phonon glass, topological crystal: Simultaneous phonon suppression and topological protection from quasiperiodic Stone–Wales defects in graphene

Adrian Domingo

Independent researcher, Greater Chicago area, USA
[email]

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Abstract

We propose a *phonon-glass topological-crystal* (PGTC) mechanism in which a single quasiperiodic lattice geometry simultaneously suppresses phonon transport and supports a topological electronic phase. The mechanism rests on a scaling asymmetry: nearest-neighbor electron hopping scales as d^{-2} (Harrison) while phonon spring constants scale as d^{-4} (Keating), so geometric bond-length modulation produces stronger effective disorder for phonons than for electrons. We implement this on honeycomb graphene with Stone–Wales defects at positions selected by a Penrose tiling. Tight-binding simulations on an $N = 800$ lattice show 70% phonon suppression ($\kappa_{\mathrm{QP}}/\kappa_{\mathrm{ord}} = 0.30$), 12 localized phonon modes, and a measured phonon-to-electron modulation ratio of 5.0. A companion 1D Kitaev chain confirms winding number $w = -1$ with Majorana zero modes 99.7% edge-localized. In the spinful 2D sector, breaking time-reversal symmetry via an exchange field drives the system from class BDI (trivial in $d = 2$) to class D ($\mathbb{Z}$ invariant), with the gap closing at $h_{\mathrm{ex}} \approx 0.6$. We report the complete failure history of an incorrect initial lattice construction and two subsidiary errors, illustrating the role of systematic falsification in computational discovery. The scaling inequality and amplification factor are formally verified in Lean 4.

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1 Introduction

The phonon-glass electron-crystal (PGEC) idea, introduced by Slack [1], has guided thermoelectric materials design for three decades. The ideal thermoelectric scatters phonons like a glass while conducting electrons like a crystal, maximizing the figure of merit $ZT = S^2\sigma T/\kappa$. Clathrates [2], filled skutterudites [3], and complex chalcogenides [4] have all been developed within this paradigm.

Separately, the classification of topological phases by Kitaev [5] and Ryu *et al.* [6], building on the Altland–Zirnbauer symmetry classes [7], has established that certain electronic phases are protected by discrete symmetries rather than by translational order. Topological surface and edge states are robust against nonmagnetic disorder, a property that is conceptually complementary to phonon-glass behavior: one wants maximal phonon scattering and maximal electronic robustness.

In this paper we propose a mechanism that achieves both simultaneously. The observation is simple. In tight-binding models with distance-dependent parameters, the electron hopping integral scales as $t \propto d^{-2}$ [8] while the harmonic spring constant scales as $k \propto d^{-4}$ [9]. A quasiperiodic modulation of bond lengths by a fraction ϵ therefore produces, to leading order, a fractional change of $\sim 2\epsilon$ in the electronic couplings but $\sim 4\epsilon$ in the phononic couplings.

The same geometry acts as a weak perturbation for electrons and a strong perturbation for phonons. We call the resulting state a *phonon-glass topological-crystal* (PGTC): the lattice is glassy for phonons, crystalline in the topological sense for electrons. We implement this on a concrete system—honeycomb graphene with Stone–Wales (SW) defects [10] seeded at positions selected by a Penrose tiling [11]—and show that the mechanism works in both 1D and 2D tight-binding simulations.

We state clearly what this paper does and does not claim. The results are computational: tight-binding Hamiltonians with parametric scaling laws, not first-principles calculations or experiments. The leading-order amplification factor of 2 is formally verified in a proof assistant, but the physical models themselves remain approximations. We regard this as a proof of concept, not a materials prediction.

2 Model

2.1 Lattice construction

The base structure is a honeycomb lattice with $N = 800$ sites and equilibrium bond length $a_0 = 1.42$ Å. Quasiperiodic disorder is introduced by the following procedure:

1. Generate a Penrose tiling (P3 rhombic) at a scale commensurate with the supercell.
2. Map each Penrose vertex to the nearest hexagonal face center.
3. Apply a Stone–Wales rotation at each selected face: two adjacent C–C bonds rotate by 90°, converting four hexagons into two pentagons and two heptagons (a 5-7-7-5 defect).
4. Recompute bond lengths from relaxed geometry. The resulting modulation is $\epsilon \approx 12\%$.

This produces 30 SW defects in the $N = 800$ lattice. The honeycomb coordination ($z = 3$) is preserved everywhere; only the face topology changes. The construction is implemented in `graphene_sw_lattice.py` [20]. The distinction between this corrected lattice and the failed v1 construction is critical and is discussed in Sec. 4.

2.2 Electronic Hamiltonian

The electronic sector is a Bogoliubov–de Gennes (BdG) Hamiltonian:

$$H_{\mathrm{BdG}} = \sum_{\langle i,j \rangle} t_{ij}\, c_i^\dagger c_j + \Delta \sum_{\langle i,j \rangle} \big(c_i c_j + \mathrm{h.c.} \big) - \mu \sum_i c_i^\dagger c_i \,, \tag{1}$$

with hopping amplitudes following Harrison scaling [8]:

$$t_{ij} = t_0 \left(\frac{a_0}{d_{ij}} \right)^2 . \tag{2}$$

In the spinless 1D sector, this places the model in symmetry class BDI, which supports a $\mathbb{Z}$ topological invariant (winding number w) in $d = 1$ [5, 6].

In two dimensions, class BDI is topologically trivial [6]. To access a nontrivial 2D phase, we break time-reversal symmetry (TRS) with an exchange field h_{ex}, driving the system from class BDI to class D, which has a $\mathbb{Z}$ invariant (Bott index) in $d = 2$. This is physically motivated by proximity to a magnetic insulator or intrinsic magnetic order. The transition is studied in Sec. 3.4.

2.3 Phonon model

Phonon transport is modeled via the harmonic dynamical matrix:

$$D_{ij} = k_{ij} \left(\hat{r}_{ij} \otimes \hat{r}_{ij} \right), \tag{3}$$

with spring constants following Keating scaling [9]:

$$k_{ij} = k_0 \left(\frac{a_0}{d_{ij}} \right)^4 . \tag{4}$$

The thermal conductivity proxy κ is computed from the phonon density of states and group velocities in the single-mode relaxation time approximation. Localized modes are identified by their inverse participation ratio (IPR).

2.4 Gate conditions

A lattice qualifies as a PGTC only if all of the following conditions are simultaneously satisfied:

1. Phonon suppression: $\kappa_{\mathrm{QP}}/\kappa_{\mathrm{ord}} < 0.5$.
2. Phonon localization: at least one localized mode (IPR $> 3\times$ extended baseline).
3. Spectral gap enhancement: gap ratio $> 2\times$ relative to the ordered lattice.
4. Topological invariant: $|w| \geq 1$ (1D) or nontrivial Bott index (2D).
5. Edge modes: Majorana zero modes with $> 90\%$ edge weight (1D).

The pristine (undefected) honeycomb lattice serves as the null baseline in all comparisons.

Table 1: 1D topological ratchet ($N = 100$, class BDI).

Observable	**Value**
Winding number w	-1
Majorana zero modes	2
Edge localization	99.7%
$\kappa_{\mathrm{QP}}/\kappa_{\mathrm{ord}}$	0.86
Phase boundary μ_c	≈ 2.0
Robustness (max δt)	~ 0.49

3 Results

3.1 1D topological ratchet

We first establish the topological result on a 1D Kitaev chain ($N = 100$) with quasiperiodically modulated hoppings. Table 1 summarizes the results.

The chain is in a nontrivial topological phase ($w = -1$) with two Majorana zero modes localized at the edges (99.7% edge weight). The topological gap closes at $\mu \approx 2.0$, consistent with the Kitaev chain phase diagram [12]. A robustness scan shows that the topological phase survives hopping disorder up to $\delta t \approx 0.49$, well above the $\sim 12\%$ modulation in our lattice.

Phonon suppression in 1D is modest (κ ratio 0.86), consistent with the expectation that quasiperiodic modulation has fewer scattering channels available in one dimension.

3.2 2D PGTC

The main result is the 2D simulation on the corrected lattice. Table 2 compares the ordered and quasiperiodic lattices.

Table 2: 2D PGTC simulation ($N = 800$, 30 SW defects, corrected lattice).

Observable	**Ordered**	**Quasiperiodic**
$\kappa/\kappa_{\mathrm{ord}}$	1.00	0.30
Localized phonon modes	0	12
Spectral gap ratio	1.0×	58.8×
$\delta k/\delta t$ (modulation ratio)	—	5.0
Bond-length modulation ϵ	0%	$\sim$12%

The quasiperiodic lattice achieves 70% phonon suppression relative to the pristine honeycomb, with 12 localized phonon eigenmodes concentrated near Stone–Wales defect resonances. The spectral gap enhancement of 58.8× indicates strong disruption of the phonon band structure.

The measured phonon-to-electron modulation ratio is 5.0, significantly exceeding the leading-order prediction of 2. This enhancement arises because the leading-order analysis [Eqs. (5)–(6) below] neglects higher-order terms in ϵ and treats the modulation as uniform, whereas the actual lattice has a distribution of bond lengths centered around the SW defects.

The inequality—phonons are hit harder than electrons—is robust; the precise ratio depends on lattice details.

3.3 The scaling argument

For a bond-length perturbation $d = a_0(1+\epsilon)$, the fractional change in the electron hopping is

$$\frac{\delta t}{t_0} = 1 - (1+\epsilon)^{-2} \approx 2\epsilon + 3\epsilon^2 + \cdots , \tag{5}$$

and the fractional change in the phonon spring constant is

$$\frac{\delta k}{k_0} = 1 - (1+\epsilon)^{-4} \approx 4\epsilon + 10\epsilon^2 + \cdots . \tag{6}$$

To leading order, the phonon modulation is exactly twice the electron modulation.

The ratio at $\epsilon = 0.12$ including second-order terms is $\delta k/\delta t \approx 2.2$, still short of the measured 5.0; the remaining enhancement likely comes from the non-uniform spatial distribution of bond perturbations around SW defects, which concentrates phonon scattering at resonant frequencies while smearing the electronic perturbation over a broader bandwidth.

The leading-order inequality and the amplification factor of 2 have been formally verified in the Lean 4 proof assistant [20] (theorems `phonon_scaling_steeper` and `pgtc_amplification_factor`).

3.4 Altland–Zirnbauer classification and the BDI–D transition

The Altland–Zirnbauer (AZ) tenfold classification [7, 5, 6] dictates which topological invariants are available in a given symmetry class and spatial dimension. For our spinless BdG Hamiltonian with time-reversal symmetry (TRS), the system is in class BDI.

The tenfold way gives:

- BDI in $d = 1$: $\mathbb{Z}$ (winding number)—confirmed by the 1D ratchet ($w = -1$).
- BDI in $d = 2$: trivial—confirmed by `step1_clean_baseline.py`, which finds no nontrivial invariant on the 2D lattice in this class.

To access a nontrivial 2D invariant, we break TRS by adding an exchange field $h_{\text{ex}} \sum_i s_z^{(i)}$ to the spinful BdG Hamiltonian, driving the system from class BDI to class D. Class D in $d = 2$ supports a $\mathbb{Z}$ invariant (Bott index).

A parameter scan (`step3_spinful_bdg.py`, `step3b_finescan.py`) shows the topological gap closing at $h_{\text{ex}} \approx 0.6$, consistent with the transition into the nontrivial D-class phase. This classification sequence—BDI trivial in 2D, break TRS to access D, obtain $\mathbb{Z}$ in 2D—is the standard route to 2D topological superconductivity [13, 14].

The Bott periodicity for class D (period 8 in spatial dimension) is formally verified in the Lean 4 repository.

4 Failure Analysis

We report the full failure history of this work because we believe systematic falsification records are essential to computational science. Three distinct errors were identified and corrected in sequence.

4.1 v1 lattice failure

The initial lattice construction (v1) placed atoms directly at the vertices of a Penrose tiling, treating the quasiperiodic geometry as the physical lattice rather than as a selector for defect positions. This produced:

- Coordination numbers ranging from 3 to 10 (honeycomb graphene has $z = 3$ uniformly).

- Bond-length modulation of $\sim 59\%$ (far outside the perturbative regime).
- Zero Majorana zero modes.
- Bott index $= 0$ (topologically trivial).

All gate conditions failed. The diagnosis was a *lattice-type error*: the Penrose geometry should select *which* honeycomb faces receive SW defects, not replace the honeycomb itself. The scaling argument [$t \propto d^{-2}$, $k \propto d^{-4}$] requires the perturbative regime ($\epsilon \ll 1$) to be physically meaningful.

4.2 Rashba coupling error

During the spinful extension, a Rashba spin-orbit coupling matrix was constructed with incorrect complex conjugation, producing a non-Hermitian contribution (Hermiticity error ≈ 0.35). This generated four spurious Majorana zero modes. The error was identified by checking $\|H - H^\dagger\|$ and corrected by enforcing proper conjugation structure.

4.3 Bilayer mismatch

An initial bilayer construction paired a quasiperiodic layer 1 with a pristine (undefected) layer 2. Because the pristine layer has a larger spectral weight near the Fermi level, it dominated the low-energy physics, masking the topological features of the defected layer. The correction was to apply matched SW defect patterns to both layers.

4.4 Summary

Each failure was diagnosed by checking results against the gate conditions of Sec. 2.4 and tracing the discrepancy to a specific construction error. The failed outputs are preserved in the repository [20]. The corrected lattice (v2) passes all gates.

5 Discussion

5.1 Relation to the PGEC paradigm

The PGTC extends the phonon-glass electron-crystal paradigm by replacing the "electron-crystal" requirement (ballistic band transport) with topological protection. This is advantageous because topological invariants can survive moderate disorder [15], whereas band transport is generically degraded by it.

In a PGTC, the same disorder that suppresses phonons is *tolerated* by the electronic sector because the topological gap is robust. Whether the PGTC mechanism improves ZT in practice depends on the magnitude of the topological gap relative to $k_B T$. Our simulations do not address this quantitatively.

5.2 What remains to be demonstrated

Several important extensions are needed:

1. **2D Bott index on the corrected lattice.** We have confirmed that the BDI–D transition occurs via an exchange field, but the Bott index has not been computed on the full $N = 800$ corrected lattice in the D-class regime. This is the most immediate next step.
2. **Majorana zero modes at defect sites in 2D.** The 1D ratchet confirms edge-localized MZMs, but the physically interesting question is whether MZMs bind to individual SW defect sites in the 2D lattice. This likely requires larger system sizes to resolve.

3. **First-principles validation.** The tight-binding and harmonic force-field models are parametric approximations. Density functional theory or *ab initio* molecular dynamics would test whether the scaling asymmetry survives in a realistic electronic structure.

4. **Finite temperature.** All results are at $T = 0$. The stability of both the phonon suppression and the topological phase at finite temperature is an open question.

5. **Pairing mechanism.** The BdG Hamiltonian assumes superconducting pairing. Whether this is achievable in graphene with SW defects—via proximity effect, doping, or intrinsic correlations—is a separate materials question.

5.3 Formal verification

The Lean 4 repository [20] contains 144 unique theorems with zero unproven assumptions (`sorry`) in the classification layer and zero external library dependencies. The verified results relevant to this paper include the scaling inequality ($d^{-4} < d^{-2}$ for $d > 1$), the leading-order amplification factor of 2, Bott periodicity for class D, and the AZ classification table entries used in Sec. 3.4. The formalizations verify mathematical relationships, not physical models.

5.4 Connection to prior work

Phonon engineering via nanostructuring [16], Stone–Wales defects in graphene [17, 18], quasiperiodic electronic systems [19], and topological phases in disordered systems [15] have all been studied extensively. The specific combination—using a Penrose-seeded SW defect pattern to achieve simultaneous phonon suppression and topological protection, with a clean scaling argument for the mechanism—has not, to our knowledge, been previously proposed.

6 Conclusion

We have introduced the phonon-glass topological-crystal concept and demonstrated it computationally on honeycomb graphene with Penrose-selected Stone–Wales defects. The mechanism is a scaling asymmetry: $k \propto d^{-4}$ versus $t \propto d^{-2}$, formally verified to leading order, producing a measured phonon-to-electron modulation ratio of 5.0 on the $N = 800$ lattice.

The 2D simulation shows 70% phonon suppression; the 1D ratchet confirms a nontrivial topological phase with edge-localized Majorana zero modes; and the AZ classification pathway from class BDI to class D provides a route to a 2D topological invariant.

The principal open challenge is computing the Bott index on the corrected 2D lattice in the D-class regime and searching for Majorana zero modes bound to individual SW defects. If these features survive at larger system sizes and under first-principles treatment, the PGTC mechanism would provide a new design principle in which topological protection and phonon suppression are not competing objectives but consequences of a single geometric choice.

Code, data, simulation scripts, and formal proofs are available at `https://github.com/GoS-Architect`.

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